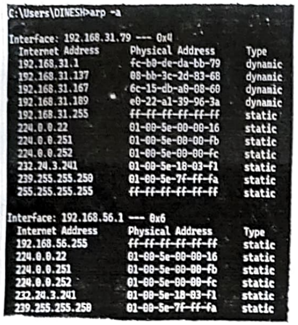
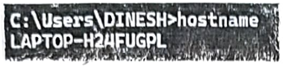
arp -a



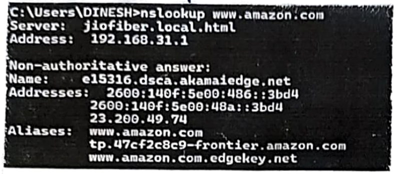
```
C:\Users\DINESH>arp -a

Interface: 192.168.31.79 --- 0x4
  Internet Address      Physical Address      Type
  192.168.31.1          fc-b0-de-da-bb-79     dynamic
  192.168.31.137        08-bb-3c-2d-83-68     dynamic
  192.168.31.167        6c-15-db-a0-08-60     dynamic
  192.168.31.189        e0-22-a1-39-96-3a     dynamic
  192.168.31.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  232.24.3.241          01-00-5e-18-03-f1     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x6
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  232.24.3.241          01-00-5e-18-03-f1     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

hostname



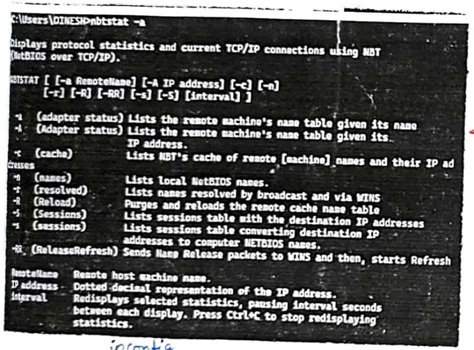
```
C:\Users\DINESH>hostname
LAPTOP-H24FUGPL
```

nslookup.



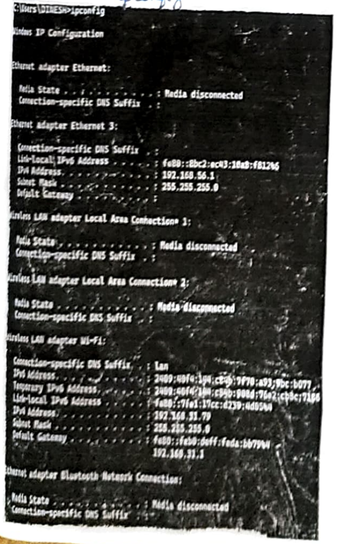
```
C:\Users\DINESH>nslookup www.amazon.com
Server:  jiofiber.local.html
Address:  192.168.31.1

Non-authoritative answer:
Name:    e15316.dsca.akamaiedge.net
Addresses:  2600:140f:5e00:486::3bd4
          2600:140f:5e00:48a::3bd4
          23.200.49.74
Aliases:  www.amazon.com
          tp.47cf2c8c9-frontier.amazon.com
          www.amazon.com.edgekey.net
```

nbtstat



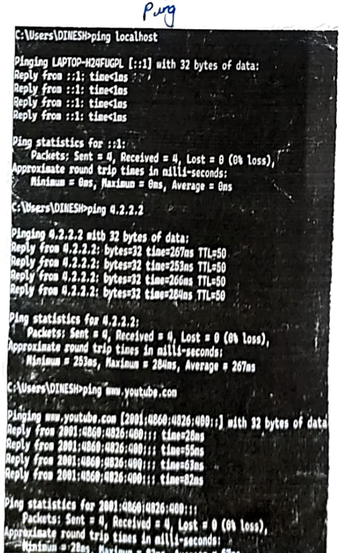
```
C:\Users\DINESH>nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP ad
dresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

ipconfig



```
C:\Users\DINESH>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::8bc2:ec43:10a8:f812%6
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : lan
   IPv6 Address. . . . . . . . . . . : 2409:40f4:104:c89b:7f70:a93:9bc:b077
   Temporary IPv6 Address. . . . . . : 2409:40f4:100:c89b:900d:7602:cb8c:7186
   Link-local IPv6 Address . . . . . : fe80::7fe1:17cc:d239:4d05%4
   IPv4 Address. . . . . . . . . . . : 192.168.31.79
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::feb0:deff:feda:bb79%4
                                       192.168.31.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

Ping



```
C:\Users\DINESH>ping localhost

Pinging LAPTOP-H24FUGPL [::1] with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms

Ping statistics for ::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\DINESH>ping 4.2.2.2

Pinging 4.2.2.2 with 32 bytes of data:
Reply from 4.2.2.2: bytes=32 time=267ms TTL=50
Reply from 4.2.2.2: bytes=32 time=253ms TTL=50
Reply from 4.2.2.2: bytes=32 time=266ms TTL=50
Reply from 4.2.2.2: bytes=32 time=284ms TTL=50

Ping statistics for 4.2.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 253ms, Maximum = 284ms, Average = 267ms

C:\Users\DINESH>ping www.youtube.com

Pinging www.youtube.com [2001:4860:4826:400::] with 32 bytes of data
Reply from 2001:4860:4826:400::: time=28ms
Reply from 2001:4860:4826:400::: time=55ms
Reply from 2001:4860:4826:400::: time=63ms
Reply from 2001:4860:4826:400::: time=82ms

Ping statistics for 2001:4860:4826:400:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 28ms,
```

16/07/2026 Ex:01 -

BASIC NETWORKING COMMANDS

Aim:

Study of various Network commands used in Linux and Windows.

Basic Networking Commands:

1. **arp-a**: ARP is a short form of address resolution protocol. It will show the IP address of your computer along with the IP address and MAC address of your router.

2. **hostname**: This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

3. **nbtstat -a**: This command helps solve problems with NetBIOS name resolution. (nbt - NETBIOS over TCP/IP).

4. **ipconfig**: This command ~~helps solve problem with NetBIOS~~ name ~~resolution~~ displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP, and type of Ethernet adapter in your system.

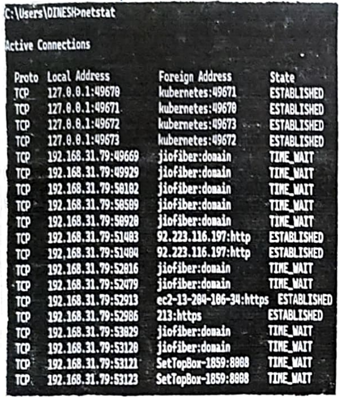
5. **nslookup**: It is a tool used to perform DNS lookups in Linux. It is used to display DNS details.

   Eg: nslookup www.google.com.

6. **Pathping**: Pathping is unique to window's and is basically a combination of the ping and Tracet commands. This command traces the route to the destination address then launches a 25 second test of each router along the way.

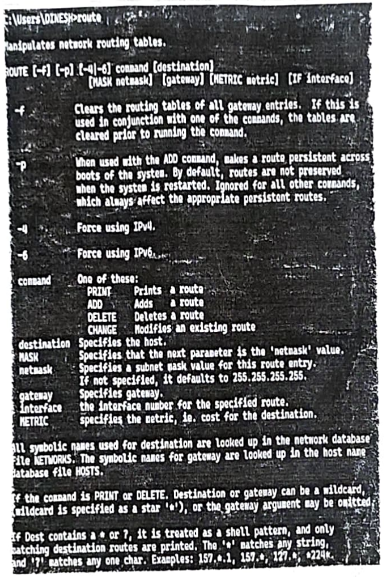
7. **Ping**: (Packet INternet Groper) command is the best way to test connectivity b/w 2 nodes.
   1. Ping hostname. (ping localhost)
   2. ping ip address. (ping 4.2.2.2)
   3. ping fully qualified domain name. (ping www.youtube.com)

netstat



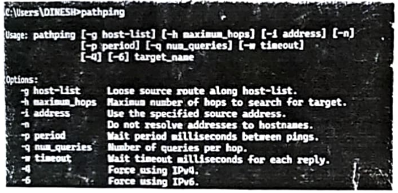
```
C:\Users\DINESH>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49670        kubernetes:49671       ESTABLISHED
  TCP    127.0.0.1:49671        kubernetes:49670       ESTABLISHED
  TCP    127.0.0.1:49672        kubernetes:49673       ESTABLISHED
  TCP    127.0.0.1:49673        kubernetes:49672       ESTABLISHED
  TCP    192.168.31.79:49669    jiofiber:domain        TIME_WAIT
  TCP    192.168.31.79:49929    jiofiber:domain        TIME_WAIT
  TCP    192.168.31.79:50102    jiofiber:domain        TIME_WAIT
  TCP    192.168.31.79:50509    jiofiber:domain        TIME_WAIT
  TCP    192.168.31.79:50920    jiofiber:domain        TIME_WAIT
  TCP    192.168.31.79:51403    92.223.116.197:http    ESTABLISHED
  TCP    192.168.31.79:51404    92.223.116.197:http    ESTABLISHED
  TCP    192.168.31.79:52016    jiofiber:domain        TIME_WAIT
  TCP    192.168.31.79:52479    jiofiber:domain        TIME_WAIT
  TCP    192.168.31.79:52913    ec2-13-204-106-34:https  ESTABLISHED
  TCP    192.168.31.79:52986    213:https              ESTABLISHED
  TCP    192.168.31.79:53029    jiofiber:domain        TIME_WAIT
  TCP    192.168.31.79:53120    jiofiber:domain        TIME_WAIT
  TCP    192.168.31.79:53121    SetTopBox-1859:8008    TIME_WAIT
  TCP    192.168.31.79:53123    SetTopBox-1859:8008    TIME_WAIT
```

route



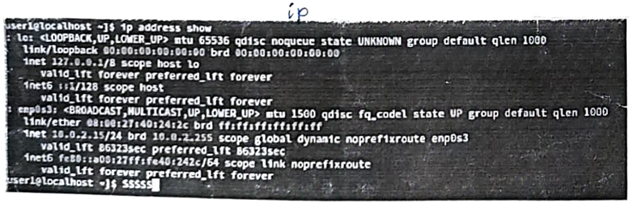
```
C:\Users\DINESH>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f           Clears the routing tables of all gateway entries.  If this is
               used in conjunction with one of the commands, the tables are
               cleared prior to running the command.

  -p           When used with the ADD command, makes a route persistent across
               boots of the system. By default, routes are not preserved
               when the system is restarted. Ignored for all other commands,
               which always affect the appropriate persistent routes.

  -4           Force using IPv4.

  -6           Force using IPv6.

  command      One of these:
                 PRINT     Prints  a route
                 ADD       Adds    a route
                 DELETE    Deletes a route
                 CHANGE    Modifies an existing route
  destination  Specifies the host.
  MASK         Specifies that the next parameter is the 'netmask' value.
  netmask      Specifies a subnet mask value for this route entry.
               If not specified, it defaults to 255.255.255.255.
  gateway      Specifies gateway.
  interface    the interface number for the specified route.
  METRIC       specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.
```

Pathping.



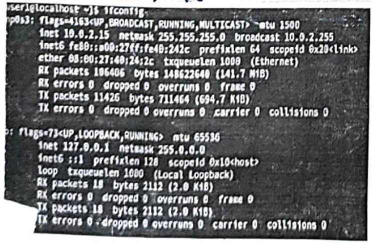
```
C:\Users\DINESH>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list     Loose source route along host-list.
    -h maximum_hops  Maximum number of hops to search for target.
    -i address       Use the specified source address.
    -n               Do not resolve addresses to hostnames.
    -p period        Wait period milliseconds between pings.
    -q num_queries   Number of queries per hop.
    -w timeout       Wait timeout milliseconds for each reply.
    -4               Force using IPv4.
    -6               Force using IPv6.
```

ip



```
user1@localhost ~]$ ip address show
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
      valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
      valid_lft forever preferred_lft forever
: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:40:24:2c brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
      valid_lft 86323sec preferred_lft 86323sec
   inet6 fe80::a00:27ff:fe40:242c/64 scope link noprefixroute
      valid_lft forever preferred_lft forever
user1@localhost ~]$ SSSSS
```

ifconfig



```
user1@localhost ~]$ ifconfig
np0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe40:242c  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:40:24:2c  txqueuelen 1000  (Ethernet)
        RX packets 106406  bytes 148622640 (141.7 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 11426  bytes 711464 (694.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 18  bytes 2112 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 18  bytes 2112 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

---

8 **Route**: route command is used to show/manipulate the IP routing table. It is primarily used to setup static routes to specific host / networks via an interface.

Linux networking commands:

1 **ip**: The ip command is one of the basic commands every administrator will need in daily work, from setting up new systems and assigning IPs to troubleshooting existing systems. The ip command can show address information, manipulate routing, plus display network various devices, interfaces and tunnels.

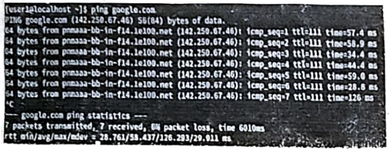
1. ip address show
2. ip address add 192.168.1.254/24 dev enp0s3
3. ip address del 192.168.1.254/24 dev enp0s3.
4. ip link set eth0 up
5. ip link set eth0 down.
6. ip link set eth0 promisc on.
7. ip route add 192.168.1.0/24 via 192.168.1.254.
8. ip route get 10.10.1.4.

2. **ifconfig**:
The ifconfig command was/is a staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has since been replaced by the ip command discussed above.

3. **mtr**:
MTR is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool combines the functionality of ping and traceroute commands.

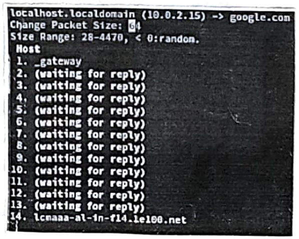
4. **tcpdump**:
The tcpdump command is designed for capturing and displaying packets.
→ You can install tcpdump with the command below
dnf install -y tcpdump.

Ping



```
[user1@localhost ~]$ ping google.com
PING google.com (142.250.67.46) 56(84) bytes of data.
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46): icmp_seq=1 ttl=111 time=57.4 ms
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46): icmp_seq=2 ttl=111 time=58.9 ms
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46): icmp_seq=3 ttl=111 time=34.4 ms
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46): icmp_seq=4 ttl=111 time=44.4 ms
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46): icmp_seq=5 ttl=111 time=59.0 ms
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46): icmp_seq=6 ttl=111 time=28.8 ms
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46): icmp_seq=7 ttl=111 time=126 ms
^C
--- google.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6010ms
rtt min/avg/max/mdev = 28.761/58.437/126.293/29.011 ms
```

tcp dump



```
localhost.localdomain (10.0.2.15) -> google.com
Change Packet Size: 64
Size Range: 28-4470, < 0:random.
 Host
 1. _gateway
 2. (waiting for reply)
 3. (waiting for reply)
 4. (waiting for reply)
 5. (waiting for reply)
 6. (waiting for reply)
 7. (waiting for reply)
 8. (waiting for reply)
 9. (waiting for reply)
10. (waiting for reply)
11. (waiting for reply)
12. (waiting for reply)
13. (waiting for reply)
14. lcmaaa-al-in-f14.1e100.net
```

5. **Ping**:

Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo Request messages. Ping is the primary TCP/IP command used to troubleshoot connectivity, reachibility and name resolution.

⇒ ping google.com.

Result: The linux and windows networking commands has been executed sucessfully.